

Smart Energy Analytics – Hourly Forecasting and Weather-Vulnerability Profiling

DALLAS Project

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Abstract

We develop a reproducible pipeline to forecast next-hour electricity consumption and quantify temperature-driven vulnerability at the building level. The pipeline builds a weather-aligned hourly dataset, creates leakage-safe time features, trains and validates tree-based models with time-aware splits, and computes temperature elasticity to segment buildings by weather sensitivity. The final artifacts include trained models, metrics, explainability plots, and a vulnerability report. This report emphasizes practical pipeline design and evaluation choices suitable for a course project, rather than proposing novel modeling techniques or theoretical advances.

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1 Introduction

Data-driven forecasting is widely used in operational settings such as energy demand management, risk assessment, and real-time decision support. In such settings, model performance alone is insufficient: practical systems must be built on leakage-free data design and time-aware evaluation, provide interpretable diagnostics, and support simple, reproducible deployment.

This project presents an end-to-end forecasting pipeline for next-hour electricity consumption using weather-aligned, hourly data from multiple regions. We engineer leakage-safe temporal and exogenous features, train tree-based models with chronological splits that reflect deployment, and analyze predictions at both global and per-building levels. Temperature sensitivity is summarized through elasticity estimates to characterize weather-related vulnerability across buildings. The resulting workflow is fully scripted and reproducible, supporting batch prediction and extension to additional regions or features.

Normalization Rationale Electricity usage is normalized on a per-building basis to remove scale effects and enable meaningful comparison across heterogeneous buildings. This normalization ensures that model performance, feature importance, and temperature elasticity reflect relative behavioral and physical responses rather than absolute load size, which is essential for cross-building analysis in a multi-region dataset.

2 Pipeline Overview

Our pipeline spans data discovery to deployment to reflect how forecasting systems are built and operated in practice:

1. Source discovery and acquisition (APIs, public archives, compliant scraping) with schema validation.
2. Unification and weather integration: align usage to hourly cadence; inner-join hourly weather; derive holidays.
3. Leakage-safe feature engineering: per-building lag_1h, lag_24h, rollmean_24h; calendar and weather drivers.
4. Time-aware splitting: chronological train/validation/test that mirror deployment.
5. Model training and evaluation: strong baselines and boosted trees with diagnostics.
6. Interpretability and vulnerability profiling: permutation importance, ICE/PDP, temperature elasticity and segments.
7. Packaging and reporting: reproducible scripts, plots, and programmatic report generation.
8. Testing and deployment: CLI for batch prediction, monitoring considerations for drift and stability.

3 Problem Definition and Scope

3.1 Problem Definition

We predict the next-hour normalized usage per building and quantify how sensitive each building's predicted usage is to temperature. Accurate short-horizon forecasts support operations (load balancing, peak management) and analytics (error attribution, risk). Temperature elasticity helps target buildings likely to be more affected by heat waves or cold spells, informing intervention

and planning. The one-hour horizon was selected as a compromise between operational relevance (short-term control and monitoring) and modeling feasibility within the scope of an hourly dataset and a course-scale project. While multi-step or longer-horizon forecasting is relevant for strategic planning, this project deliberately focuses on a one-hour horizon. This choice reduces compounding forecast error, simplifies leakage control, and aligns with operational monitoring use cases, making it appropriate for a course-scale implementation with hourly data.

3.2 Scope and Data Acquisition Strategy

We focus on hourly resolution across multiple regions, retaining only hours with weather to avoid bias when analyzing temperature effects. Sources are selected for schema consistency and time semantics. We prefer features that reflect real operational signals (time-of-day, recent usage, weather), and evaluate models with chronological splits that emulate deployment. Below we summarize typical use cases that ground our scope:

- **Operational planning:** next-hour forecasts support load balancing and peak management.
- **Targeted interventions:** high-sensitivity buildings (elasticity) for demand response and outreach.
- **Policy evaluation:** track the distribution of elasticity across time and customer segments.

4 Data Acquisition and Dataset Characteristics

4.1 Data Source and Collection Methodology

We combined two public sources (Ausgrid NSW and London LCL) and standardized them to hourly alignment, merging per-region weather (apparent temperature, precipitation, day/night). We retained only rows with weather present to avoid exogenous imputations. Identifiers are anonymized. The unified dataset serves as the single source of truth for feature engineering and chronological evaluation.

In practice, we first surveyed public archives and APIs for smart-meter or feeder-level usage data across regions. Where APIs were unavailable, we used compliant scraping to collect candidate datasets and validated schemas for usage fields, time resolution, and coverage. Several sources were excluded due to missing usage fields or incompatible granularities. We converged on Ausgrid (NSW, AU) and London LCL as consistent sources. Weather and holiday signals were integrated per region: hourly apparent temperature, precipitation, and day/night flags (temperature normalized), with public holidays derived programmatically (e.g., AU NSW, GB calendars). We kept only rows with weather present to avoid imputing exogenous drivers, preserving integrity for temperature-sensitivity analyses. This filtering reduces sample size but avoids introducing artificial assumptions about exogenous drivers, which is particularly important when interpreting temperature sensitivity.

Item	Value
Rows	74498698
Buildings	3835
Time range	2007-01-02 00:00:00 to 2022-04-30 22:00:00
Train end	2014-01-26 15:00:00
Validation end	2017-06-17 05:00:00
Test period	> 2017-06-17 05:00:00

Table 1: Dataset scope and chronological cutoffs.

4.2 Technical Challenges and Solutions

- Heterogeneous schemas across sources (CSV vs Parquet) were normalized to a common long format (building-hour rows). - Weather alignment: hourly join with strict inner matches; rows without weather removed to preserve interpretability in temperature analyses. - Leakage controls: lags/rolling computed per building with shift operations; chronological splits prevent look-ahead. - Scale and performance: chunked reads and sampling for EDA; compact figures and tables generated from representative samples.

4.3 Feature Engineering

Group	Features
Calendar	hour, day_of_week, month, season, is_weekend, is_holiday
Dynamics	lag_1h, lag_24h, rollmean_24h (leakage-safe)
Weather	apparent_temperature_norm, precipitation, is_day
Target	y_next (next-hour usage per building)

Table 2: Feature groups used in modeling.

Weather Normalization Apparent temperature is normalized to the $[0, 1]$ range within each region to stabilize model training and enable consistent interpretation of temperature effects. This scaling allows temperature elasticity to be compared across buildings without conflating absolute climate differences between regions.

5 Experimental Design and Splits

We estimate time cutoffs via reservoir sampling of timestamps to approximate 80%/10%/10% train/val/test boundaries.

Random splits were deliberately avoided, as they would introduce temporal leakage and lead to overly optimistic estimates that do not reflect deployment conditions. MAE is used as the primary metric because it is robust to outliers and provides a stable measure of typical prediction error. RMSE is reported as a secondary metric to highlight sensitivity to occasional large deviations, which may be operationally costly.

Train end=2014-01-26 15:00:00, **Validation end**=2017-06-17 05:00:00; Test strictly later.
Metrics: MAE (primary), RMSE (secondary).

6 Model Development, Training, and Evaluation

6.1 Feature Selection, Preprocessing, and Initial Analysis

We curated features that reflect operational reality and avoid leakage: per-building lag_1h and lag_24h capture persistence and daily cycle; rollmean_24h summarizes recent regime; calendar features anchor seasonality and hour-level structure; weather features (normalized apparent temperature, precipitation, is_day) act as exogenous drivers. Preprocessing is minimal by design (trees are scale-robust), focusing effort on correct time alignment and splits.

6.2 Baselines and Primary Model

- Naive: $\hat{y}(t) = y(t - 1)$
- RandomForestRegressor (stable baseline)
- HistGradientBoostingRegressor (primary)

Training uses large sampled subsets per split to bound memory while preserving time ordering.

7 Key Findings

- Boosted trees outperform Naive by 45.5% MAE (val) and 37.7% MAE (test); RMSE improvements: 46.4% (val), 34.5% (test).
- Temperature, hour-of-day, and recent-usage dynamics rank highly by permutation importance.
- Temperature elasticity median = 0.0108; segments: Low=48, Moderate=94, High=48.

8 Results

8.1 Model Performance Evaluation

We evaluate models using MAE (robust to outliers) and RMSE (penalizes large errors) on both validation and hold-out test periods, computed strictly after chronological cutoffs to emulate deployment. Reporting both metrics helps separate median-case fidelity from tail-risk behavior, which is important for operational decision making where occasional spikes can incur high costs.

Model	MAE (val)	RMSE (val)	MAE (test)	RMSE (test)
Naive	0.0384	0.0535	0.0478	0.0625
Random Forest	0.0239	0.0332	0.0330	0.0453
HistGBR	0.0209	0.0287	0.0298	0.0410

Table 3: Overall accuracy across models (val/test). Primary model: HistGBR.

8.2 Feature Importance (Validation)

Permutation importance highlights `lag_1h`, `hour`, `lag_24h`, `rollmean_24h`, `day_of_week` as leading drivers. Higher values indicate stronger contribution to reducing validation MAE; the prominence of temperature and recent usage aligns with expected physical and behavioral effects.

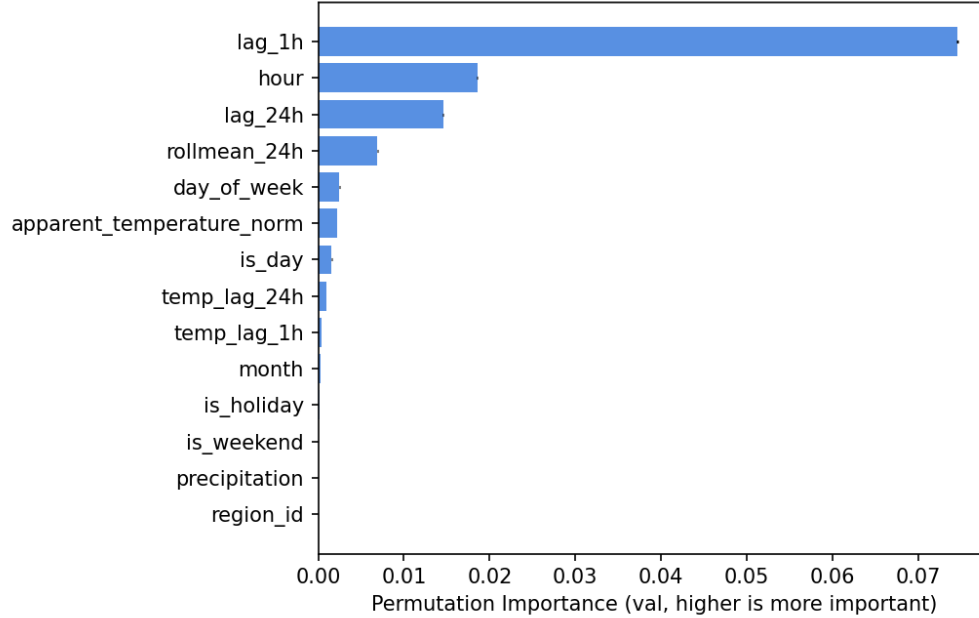


Figure 1: Permutation importance (validation) for HistGBR. Dominant features confirm the role of temperature and daily dynamics.

Top features (if computed):

Feature	Importance (mean)	Std
lag_1h	0.075	0.000
hour	0.019	0.000
lag_24h	0.015	0.000
rollmean_24h	0.007	0.000
day_of_week	0.002	0.000
apparent_temperature_norm	0.002	0.000
is_day	0.002	0.000
temp_lag_24h	0.001	0.000
temp_lag_1h	0.000	0.000
month	0.000	0.000
is_holiday	0.000	0.000
is_weekend	0.000	0.000
precipitation	0.000	0.000
region_id	0.000	0.000

Table 4: Top features by permutation importance (validation).

8.3 Fit Quality and Error Patterns

The scatter below should concentrate around the diagonal if the model is well calibrated; dispersion at higher usage levels reflects the heavier-tailed regime where short spikes are harder to capture.

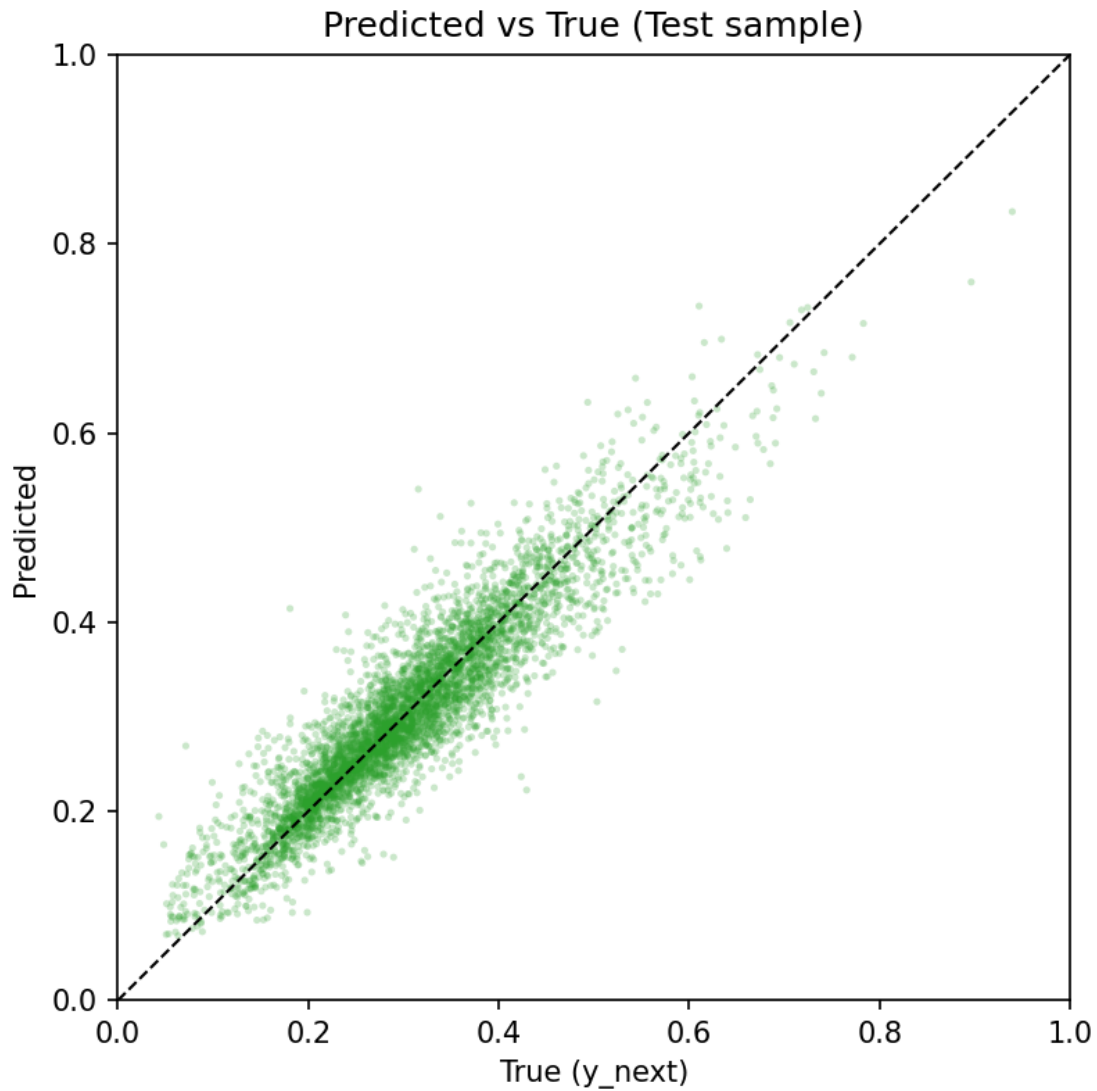


Figure 2: Predicted vs True (test sample). Points near the diagonal indicate good calibration; dispersion at high usage reflects spiky demand that remains challenging.

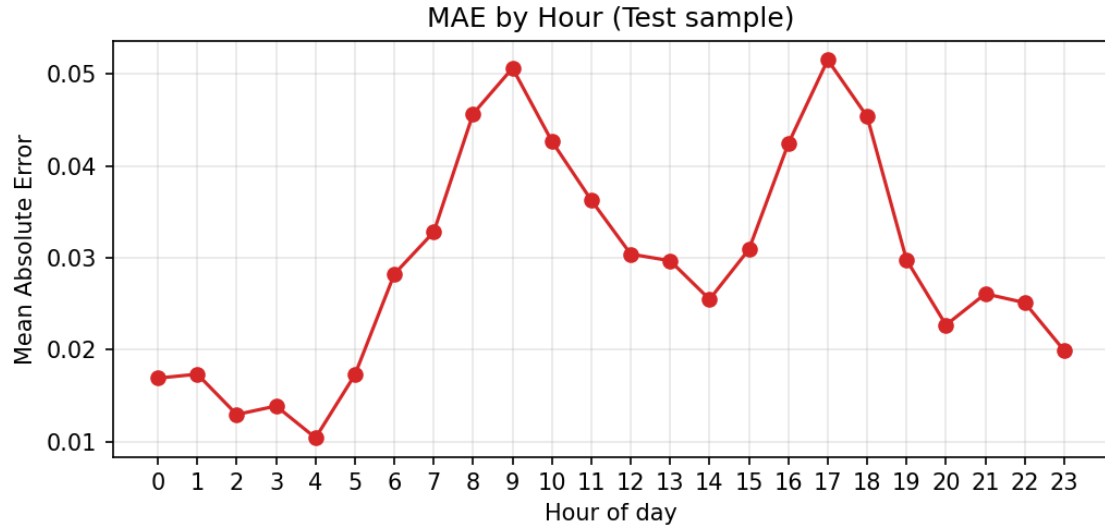


Figure 3: MAE by hour of day on the test sample. Elevated errors around transition hours (morning/evening) indicate regime changes are hardest to predict.

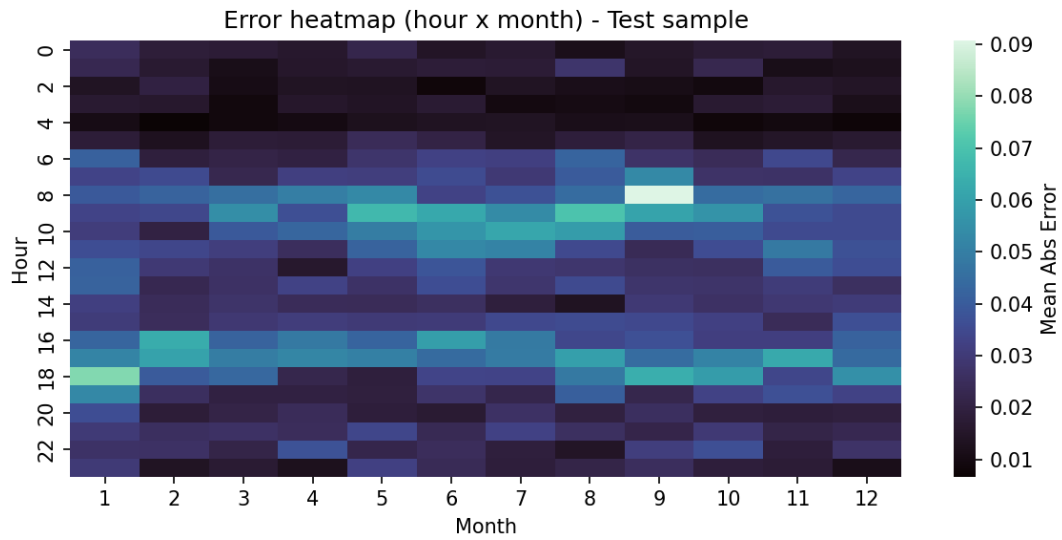


Figure 4: Error heatmap by hour and month (test sample). Warm bands reveal seasonal-hourly regimes where performance degrades, guiding targeted feature or segment refinements.

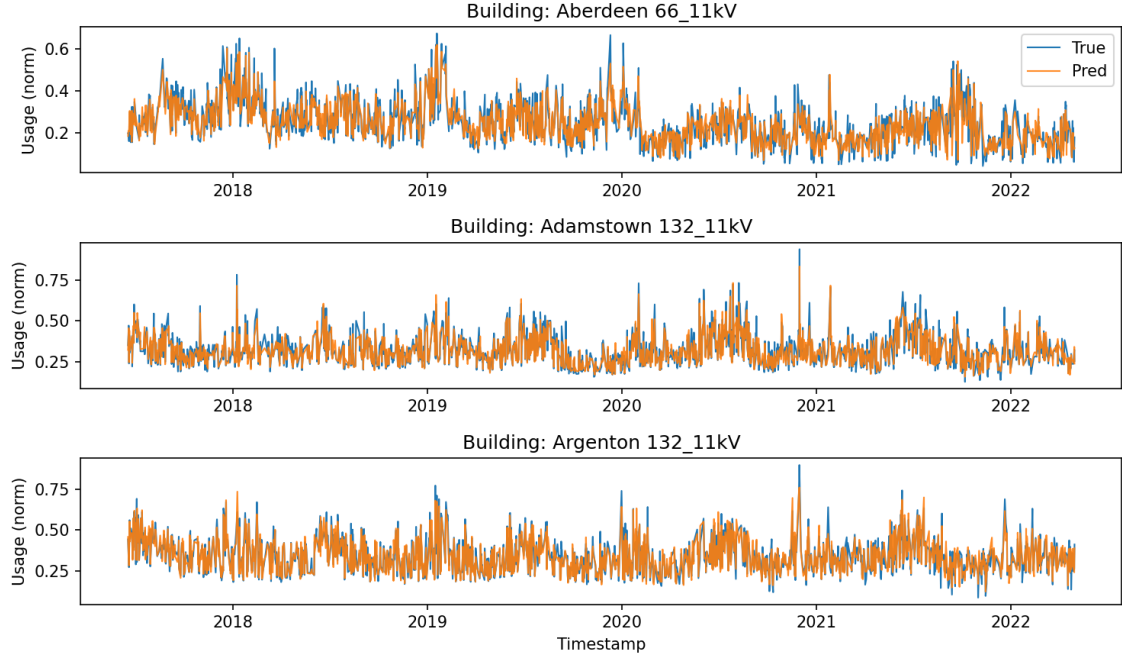


Figure 5: Example time series: predicted vs actual for top buildings in sample. The model tracks levels and turning points with small lag except during sharp spikes.

Summary of prediction behavior Across these diagnostics, the model demonstrates strong calibration and the ability to track short-term demand dynamics under typical conditions. Remaining errors are concentrated in spike-like regimes and transition periods, which are inherently difficult to capture with short-horizon autoregressive features. This observation motivates the subsequent interpretability and vulnerability analysis.

8.4 Per-Building Test MAE Summary

$n=3$, $\min=0.0273$, $Q1=0.0282$, $\text{median}=0.0291$, $Q3=0.0310$, $\max=0.0330$.

9 Model Interpretability with SHAP Analysis

We interpret the model using global importance and response analyses. While SHAP provides additive attributions, for this project we adopt permutation importance (global) and ICE/PDP (local and global response) that convey similar directional insights for tree ensembles. We inspect how predictions change with temperature and time features, and summarize building-level sensitivity via elasticity. Given the scale of the dataset and the educational scope of the project, PDP and ICE plots were preferred over full SHAP value computation to balance interpretability, computational cost, and clarity of presentation.

9.1 Method

We compute per-building temperature elasticity by varying `apparent_temperature_norm` on a fixed grid $[0,1]$ for sampled hours (test period), predicting with the trained model, and fitting a line $dy/dtemp$ across the ICE curve. The building elasticity is the median slope. It is important to note that the estimated temperature elasticity reflects model-implied sensitivity under observed conditions and does not establish causal effects. The metric should therefore be interpreted as a comparative vulnerability indicator rather than a causal coefficient. Intuitively, a higher elasticity indicates that predicted demand responds more strongly to temperature changes, making such

buildings more exposed to weather extremes. In this sense, elasticity serves as a practical proxy for temperature-driven vulnerability.

9.2 Segments and Distribution

Segments by quantiles: Low=48, Moderate=94, High=48. Median elasticity=0.0108; Q1=0.0076, Q3=0.0149 (scored 190/190 buildings).

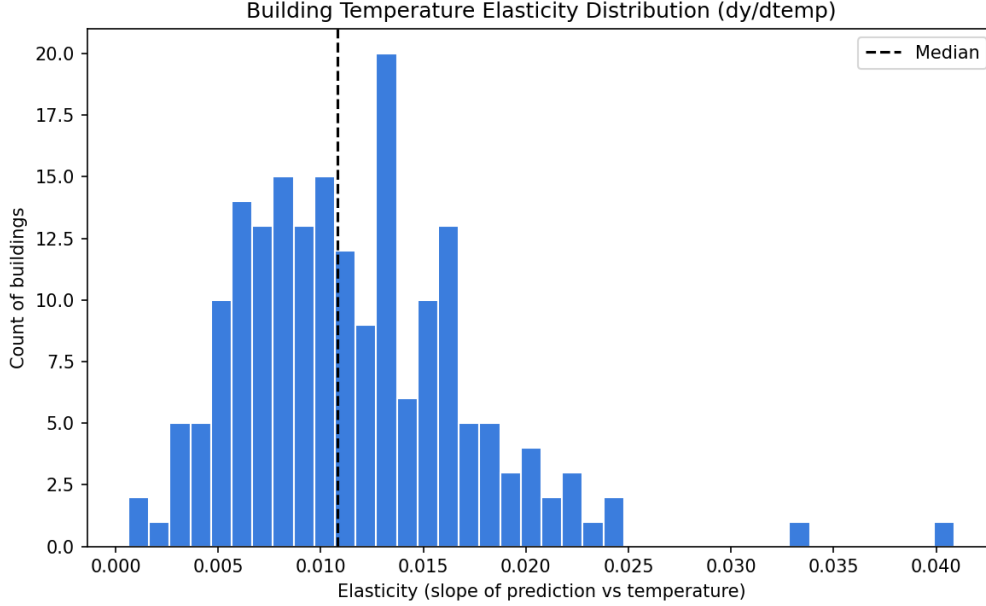


Figure 6: Distribution of building temperature elasticity (dy/dtemp).

9.3 Top/Bottom Temperature Sensitivity (Test Period)

Top 12 High-Sensitivity Buildings	
Building	Elasticity
Mt Hutton 33_11kV	0.0409
Port Botany 33_11kV	0.0329
Somersby 132_11kV	0.0247
Darling Harbour 132_11kV	0.0238
Paddington 33_11kV	0.0237
St Peters 132_11kV	0.0225
Revesby 132_11kV	0.0219
Berowra 132_11kV	0.0218
Hornsby 132_11kV	0.0216
Lisarow 33_11kV	0.0210
Kirrawee 132_11kV	0.0207
Kingsford 132_11kV	0.0207

Table 5: Highest positive elasticity (dy/dtemp).

Top 12 Low-Sensitivity Buildings	
Building	Elasticity
Crows Nest 33_11kV	0.0006
Tomago 33_11kV	0.0012
Tomalpin 33_11kV	0.0021
Milperra 132_11kV	0.0028
Tighes Hill 33_11kV	0.0029
Mitchells Flat 66_11kV	0.0031
Engadine 33_11kV	0.0033
Avoca 66_11kV	0.0036
Toronto 33_11kV	0.0037
Broadmeadow 33_11kV	0.0038
Carrington 33_11kV	0.0039
City East 33_11kV	0.0043

Table 6: Lowest elasticity (dy/dtemp).

10 Exploratory Data Analysis (EDA)

10.1 How to read the EDA figures To avoid presenting figures without interpretation, we group the diagnostics into four purposes: (i) data integrity (missingness), (ii) demand regimes (distribution and seasonality), (iii) feature relationships (correlation and partial dependence), and (iv) failure modes (error slices by hour, season, and temperature). For each group, we state the main takeaway and how it motivates either model design (features/splits) or vulnerability interpretation (elasticity).

We first validate the data mechanics (missingness, distributions) and then study relationships. The goal is to understand regimes where the model must perform well and where risk is elevated. The EDA serves not to exhaustively analyze all patterns, but to verify data integrity, reveal dominant regimes, and ensure that the subsequent modeling assumptions are reasonable.

10.1 Summary of EDA Findings

- Usage distributions exhibit expected seasonality and diurnal patterns; weekend/holiday behavior differs modestly by region.
- Error analysis shows the hardest hours are transitions (morning/evening) and hotter temperature bins in some regions.
- Correlation heatmaps suggest low multicollinearity among engineered signals, a favorable setting for tree models.
- Usage vs temperature (normalized) shows expected nonlinearity and heteroscedasticity.
- Hour-of-day error patterns indicate highest difficulty during regime transitions (early morning/evening).
- Seasonality captured via lag_24h and monthly effects.

10.2 Extended EDA and Diagnostics (Selected Figures)

The following figures were selected to validate data mechanics and illuminate model-relevant structure. Missingness views ensure inputs are trustworthy; distributional plots reveal heterogeneity across regions and seasons; temporal heatmaps capture diurnal and monthly regimes

the model must learn; correlation heatmaps check for multicollinearity that could impair generalization; residual analyses, calibration, and PDP/ICE plots connect model behavior back to physical drivers. Together, these diagnostics justify feature choices and highlight risks that guide improvement.

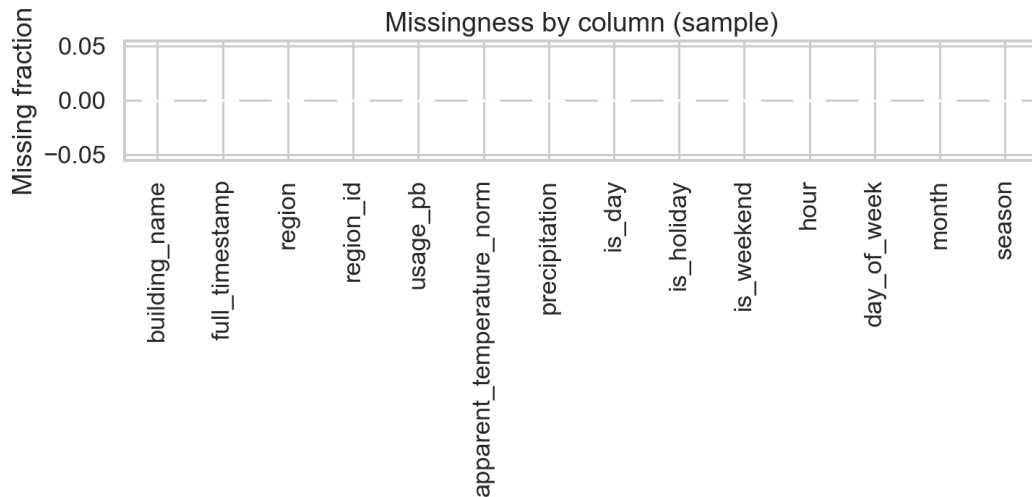


Figure 7: Missingness by column (sample).

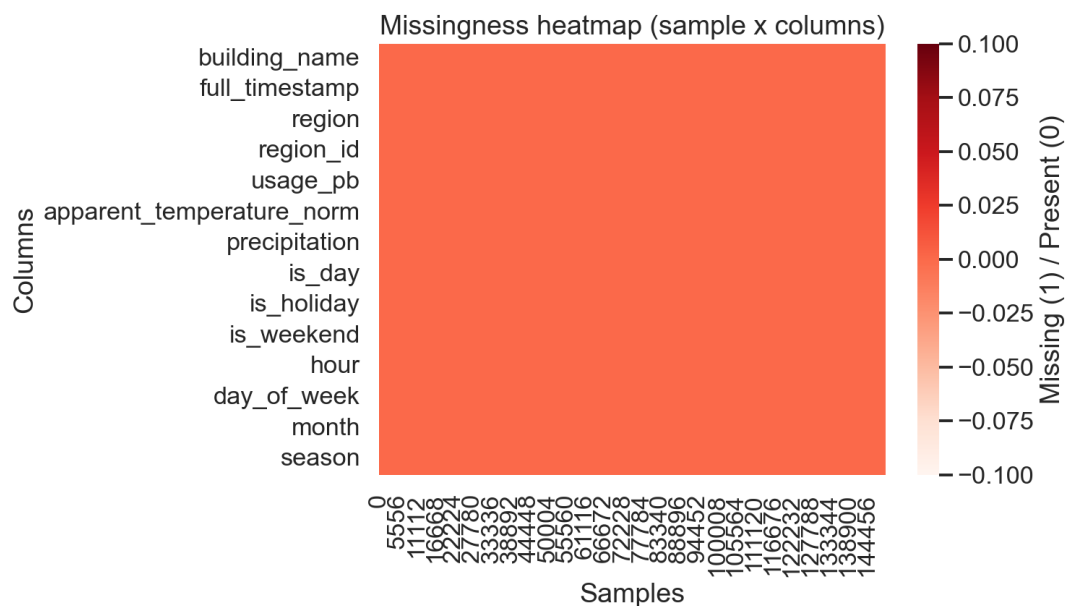


Figure 8: Missingness heatmap (sample x columns).

Takeaway (Missingness) The missingness diagnostics confirm that the modeling table is largely complete after the weather-aligned inner join, reducing the need for imputation that could bias interpretability. This supports the validity of downstream temperature-sensitivity analysis, where artificial filling of exogenous variables would otherwise distort elasticity estimates.

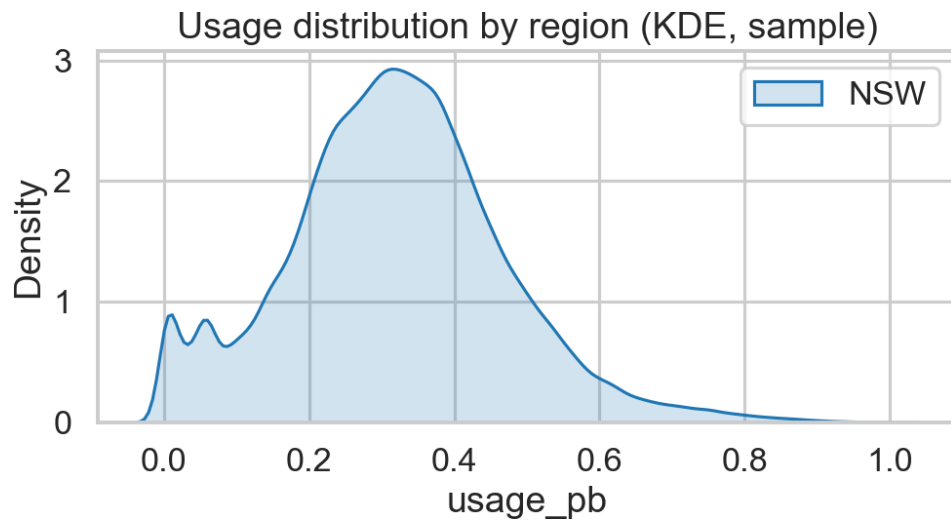


Figure 9: Usage distribution by region (KDE).

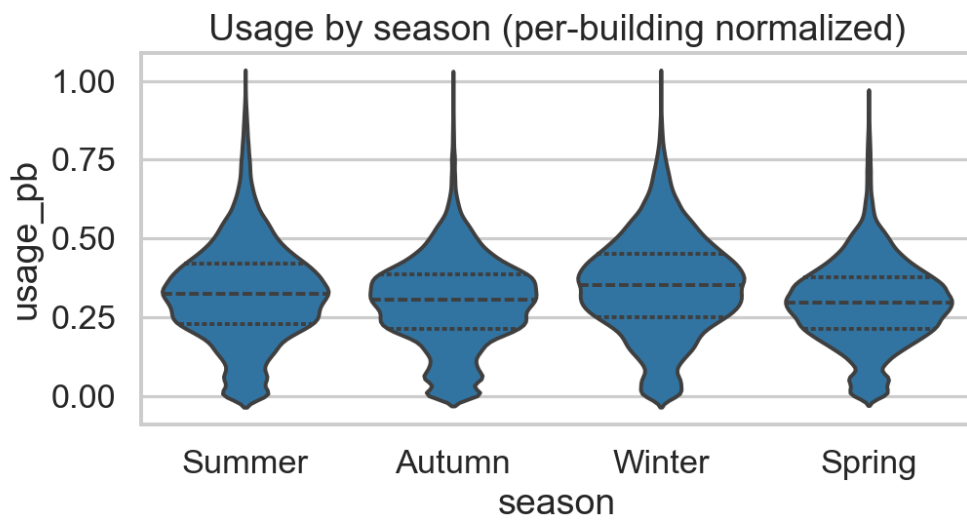


Figure 10: Usage by season (per-building normalized).

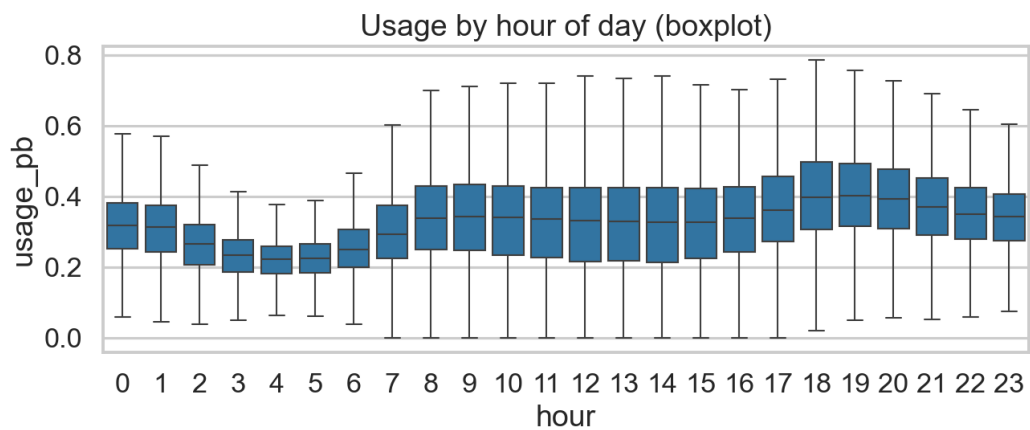


Figure 11: Usage by hour of day (boxplot).

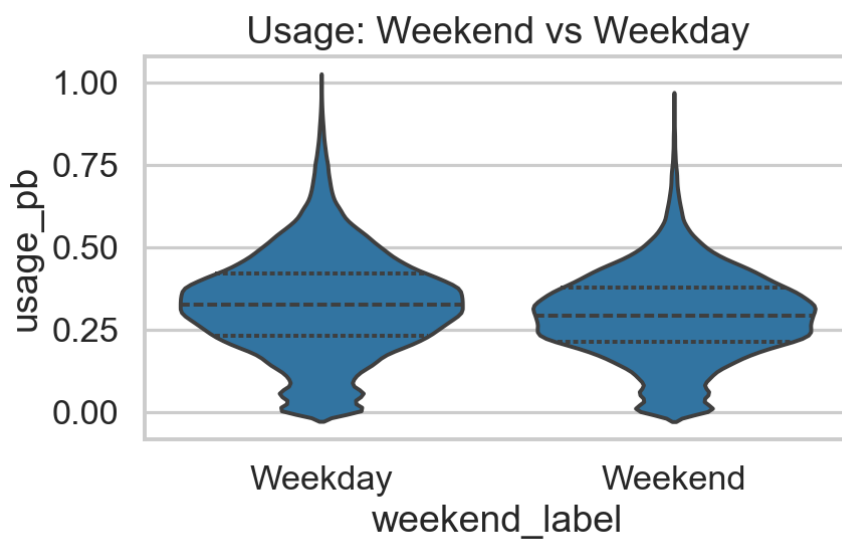


Figure 12: Usage: Weekend vs Weekday.

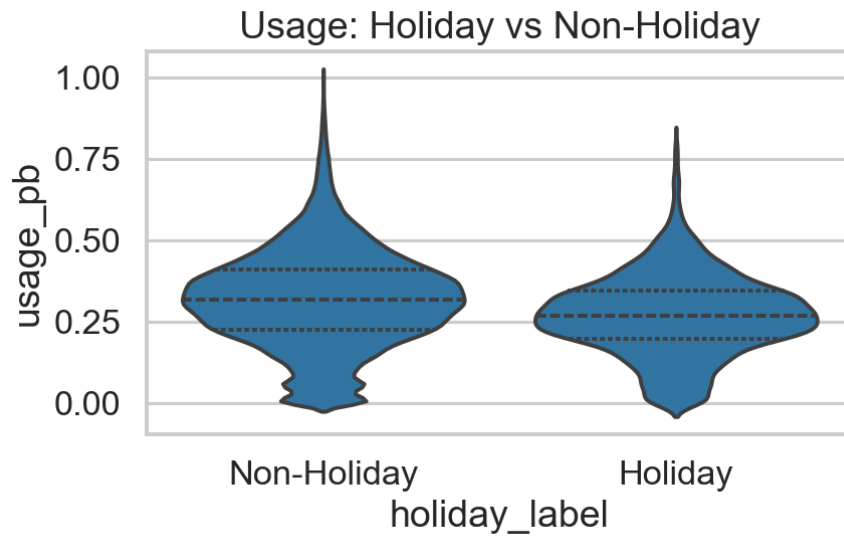


Figure 13: Usage: Holiday vs Non-Holiday.

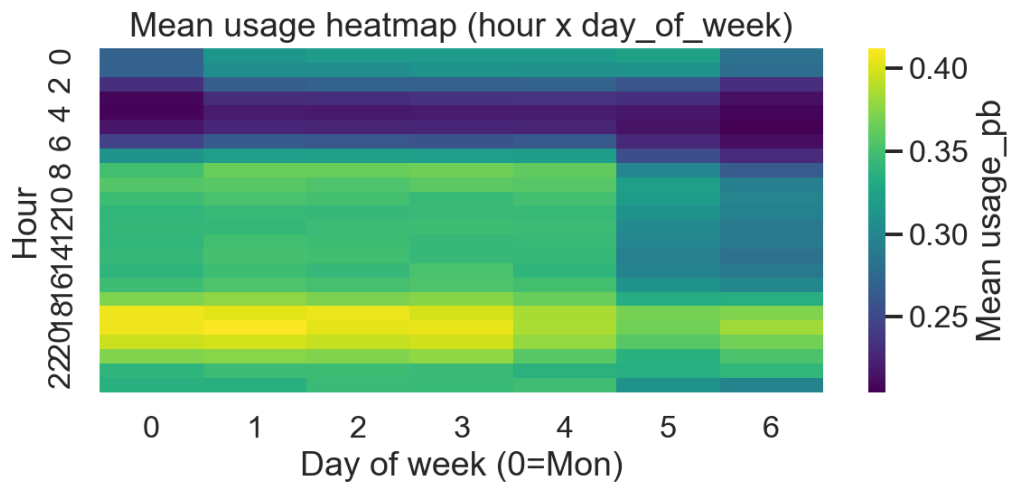


Figure 14: Mean usage heatmap (hour x day_of_week).

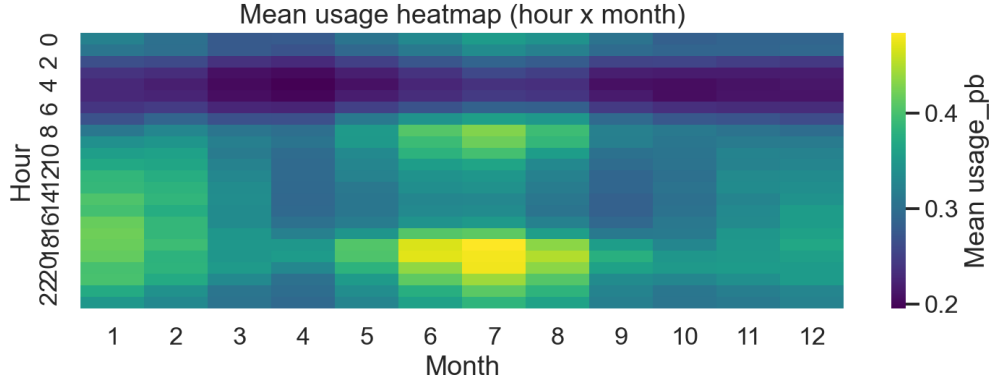


Figure 15: Mean usage heatmap (hour x month).

Takeaway (Demand regimes) The distribution and seasonality plots show clear diurnal structure and moderate seasonal shifts, indicating that hour-of-day and lagged usage features should be strong predictors. Weekend/holiday effects appear present but smaller, suggesting that calendar indicators are useful as secondary signals rather than the primary drivers of next-hour demand.

10.3 Correlation Analysis

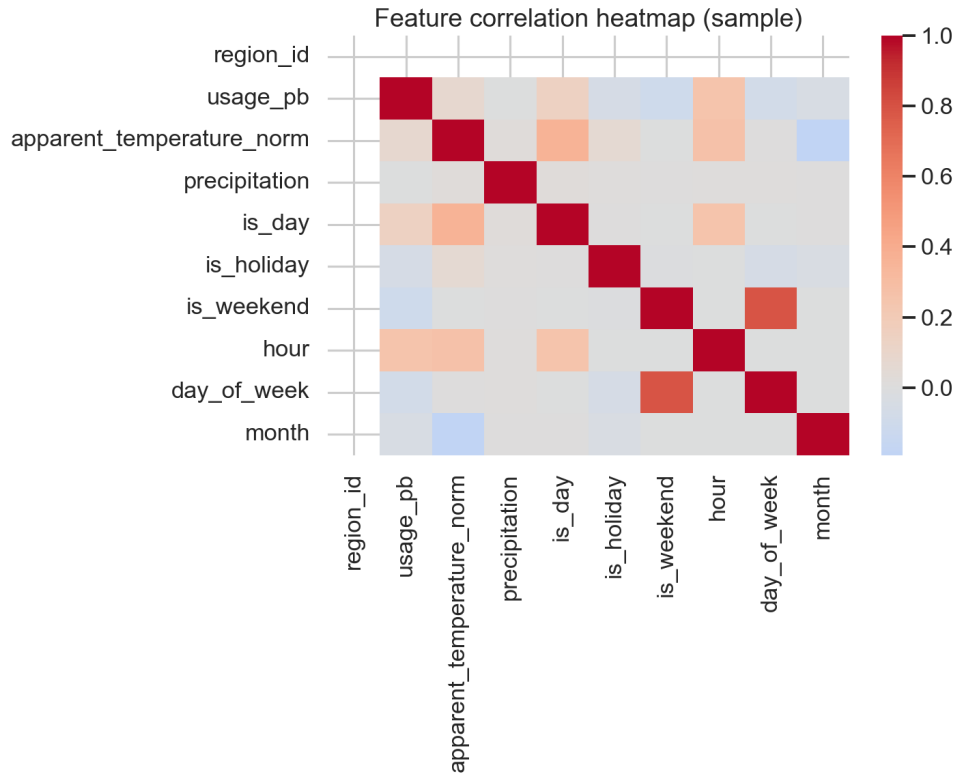


Figure 16: Feature correlation heatmap (sample).

Takeaway (Feature relationships) The correlation heatmap suggests limited multicollinearity among the engineered features, which is favorable for tree-based models and for interpreting

feature effects via PDP/ICE. This reduces the risk that apparent importance is driven purely by redundant proxies.

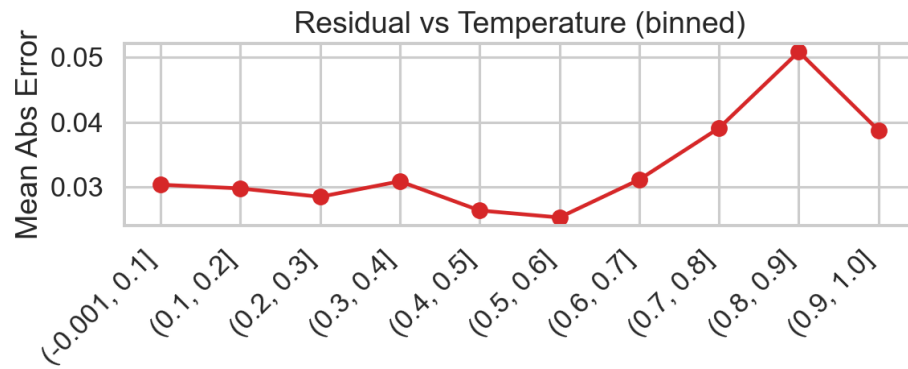


Figure 17: Residual vs Temperature (binned MAE).

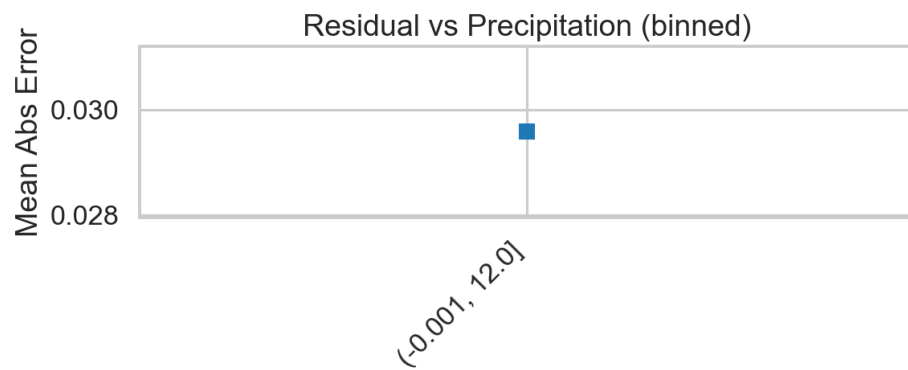


Figure 18: Residual vs Precipitation (binned MAE).

Takeaway (Where the model fails) Errors increase in specific temperature bins and during transition hours, implying that a single global model with limited exogenous drivers may underfit rare, spike-like regimes. This motivates future work on richer exogenous inputs and segment-aware modeling for high-risk periods.

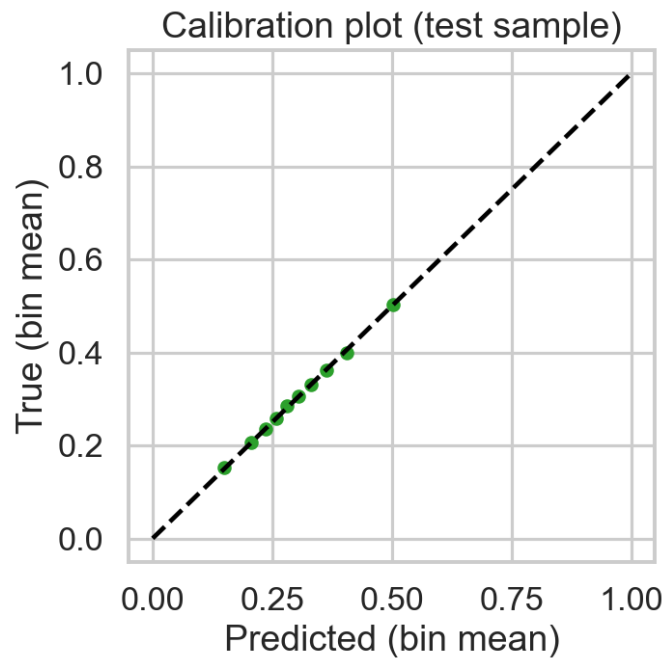


Figure 19: Calibration: predicted deciles vs true mean.

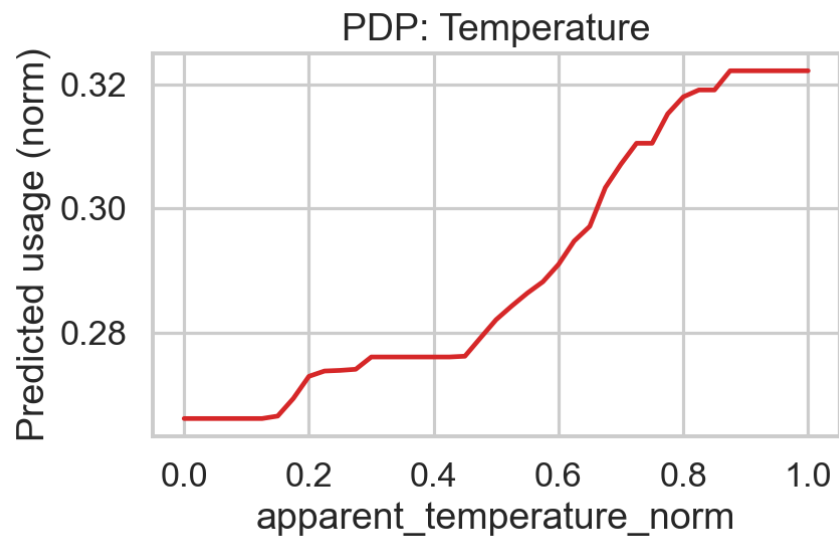


Figure 20: PDP: Temperature effect on predicted usage.

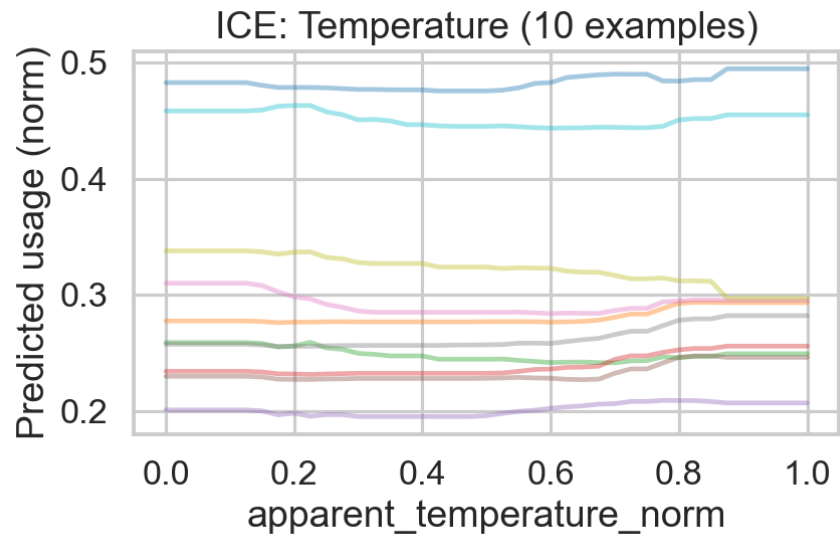


Figure 21: ICE: Temperature (10 examples).

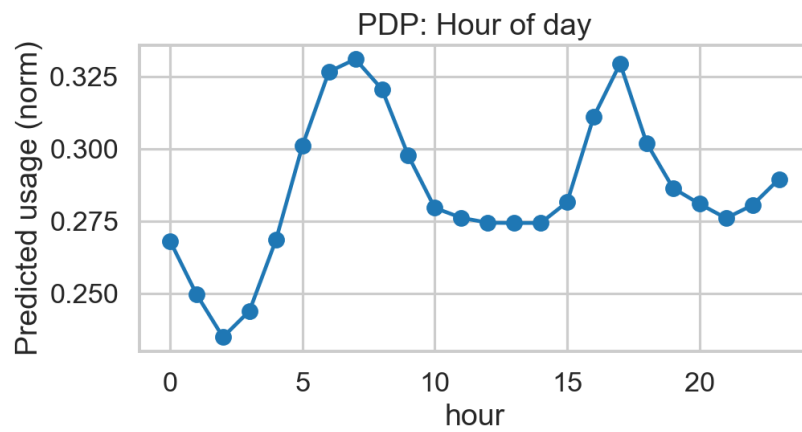


Figure 22: PDP: Hour of day.

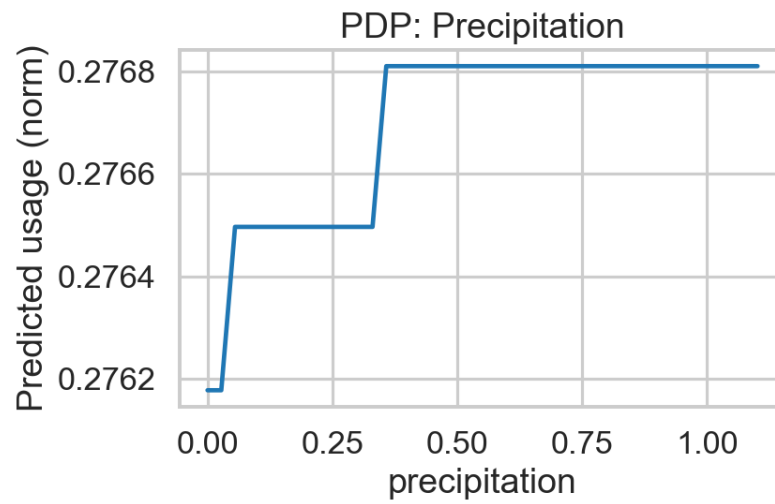


Figure 23: PDP: Precipitation.

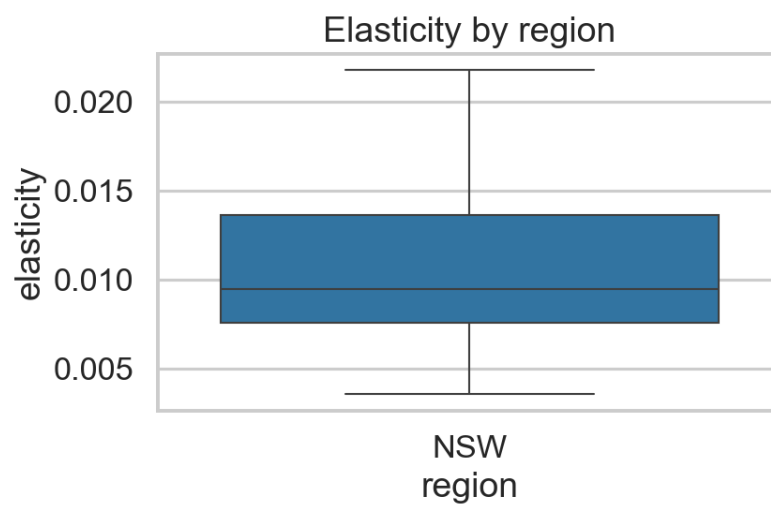


Figure 24: Elasticity by region (boxplot).

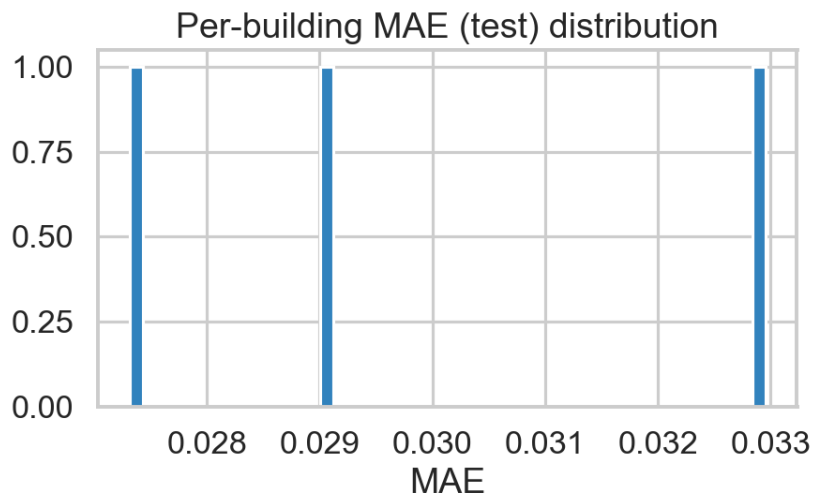


Figure 25: Per-building MAE distribution (test).

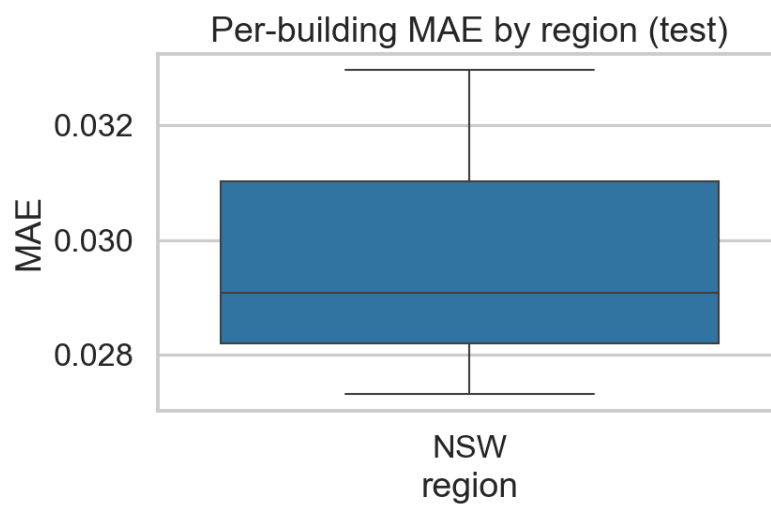


Figure 26: Per-building MAE by region (test).

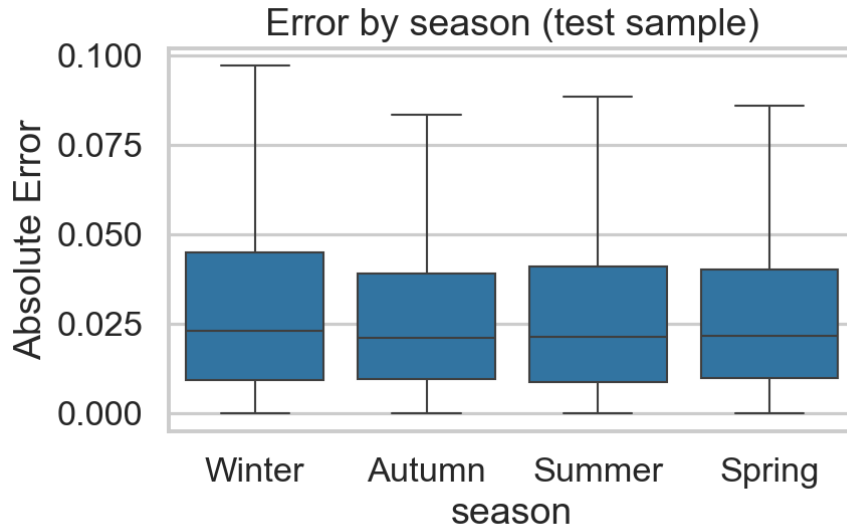


Figure 27: Error by season (test sample).

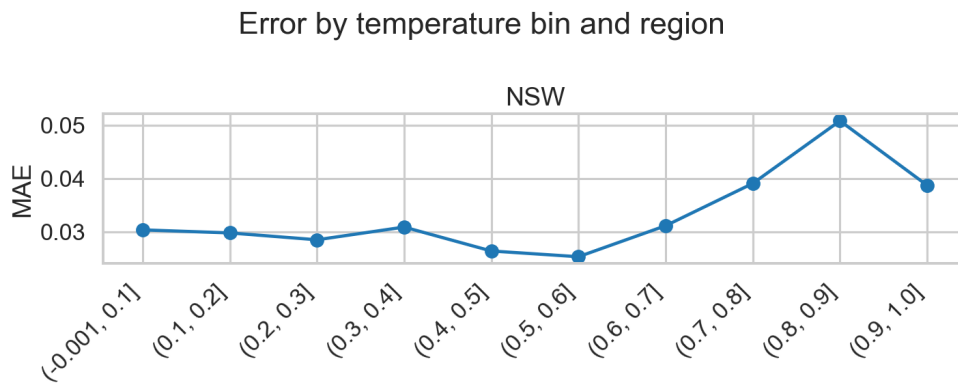


Figure 28: Error by temperature bin and region. Temperature regimes differ by region; higher bins align with larger errors in hotter periods.

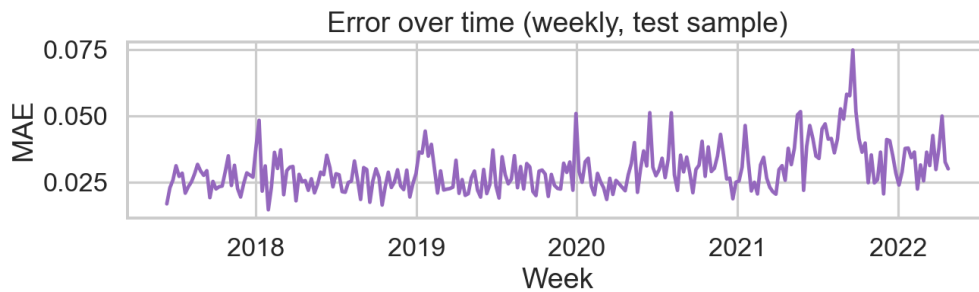


Figure 29: Error over time (weekly MAE, test sample). Stable weekly MAE with mild variation suggests good temporal generalization.

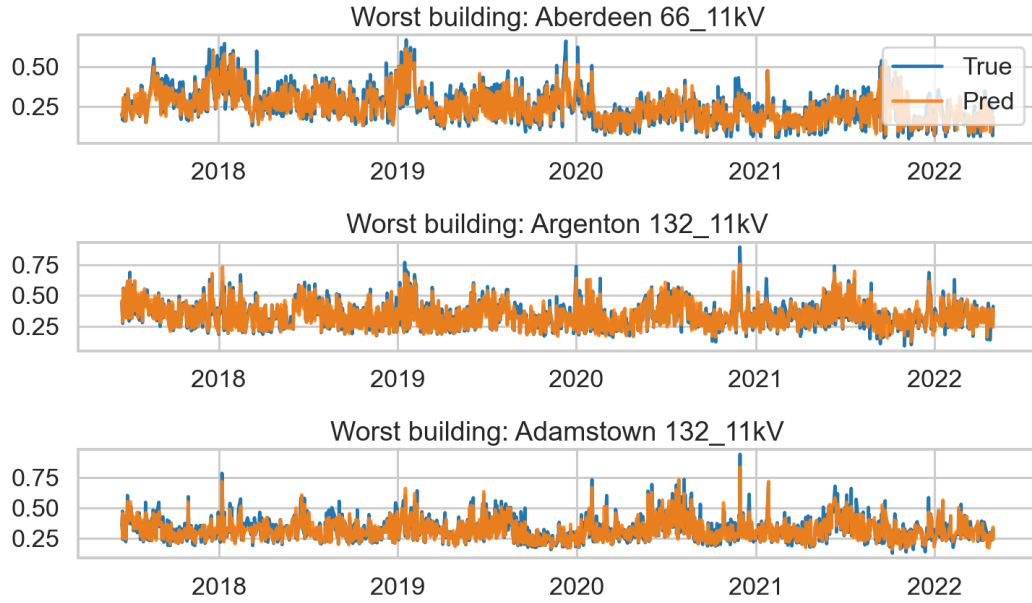


Figure 30: Worst buildings (by MAE): predicted vs actual time series.

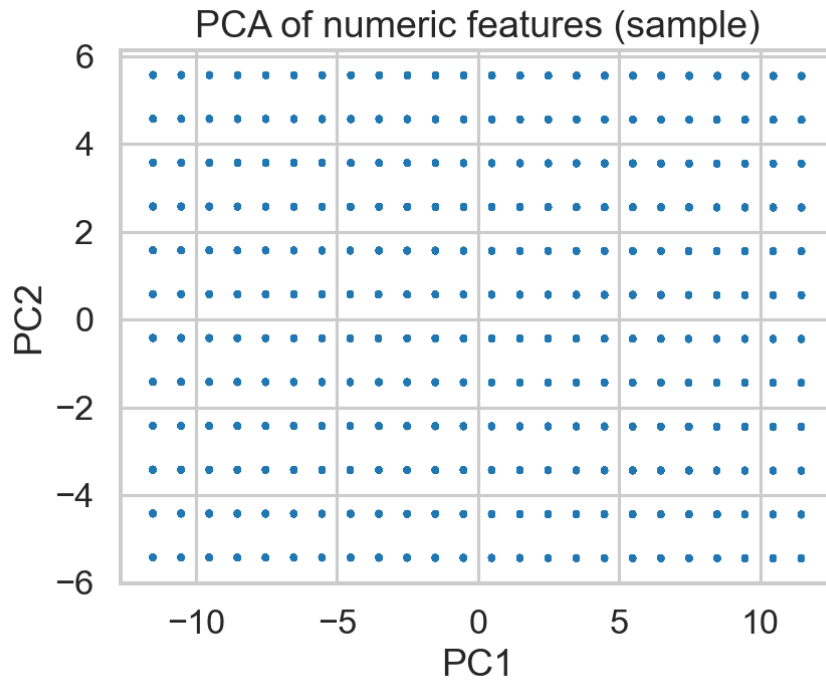


Figure 31: PCA of numeric features (sample).

Why PCA and clustering are included We include PCA and simple clustering not as primary modeling tools, but as sanity checks on heterogeneity: if buildings form distinct usage-profile groups, segment-specific models or elasticity-based stratification may be justified.

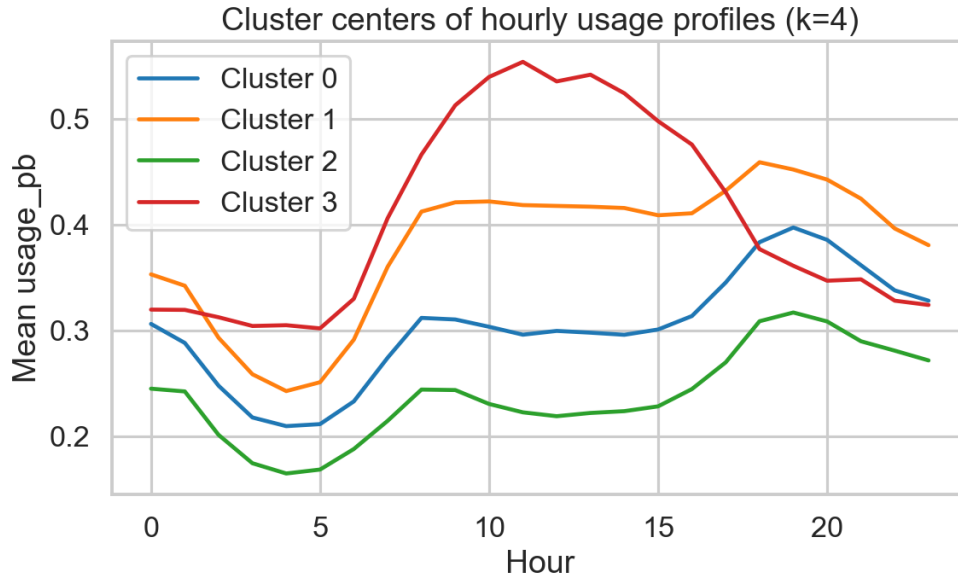


Figure 32: Cluster centers of hourly usage profiles (k=4).

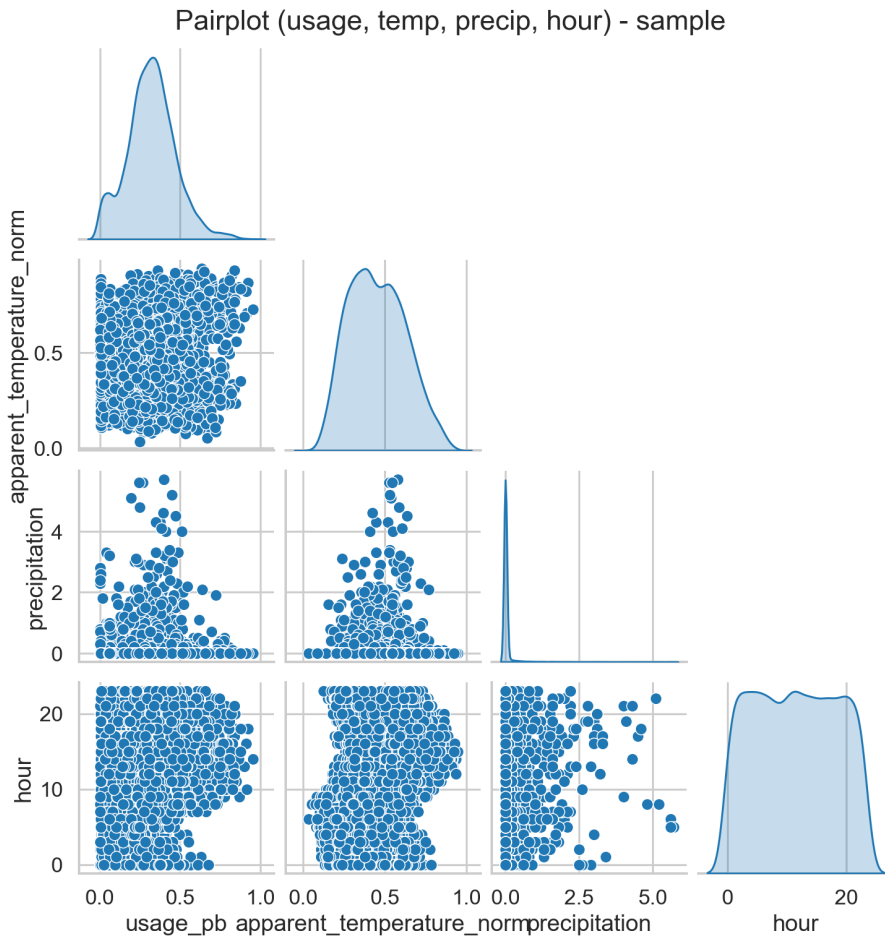


Figure 33: Pairplot (usage, temp, precip, hour) on a sample.

Although these analyses are exploratory, they confirm that usage patterns exhibit structured

heterogeneity rather than forming a single homogeneous group. This supports the interpretation of elasticity differences as reflecting meaningful behavioral or physical variation across buildings.

Implications for modeling The observed heterogeneity in usage profiles suggests that a single global model may mask building-specific response patterns. While this project adopts a unified model for simplicity, the results support future exploration of segment-aware forecasting or elasticity-driven stratification.

10.4 Model Selection and Training

We compared a naive persistence baseline, Random Forest (variance reduction, stable baseline), and HistGradientBoosting (tabular boosting with strong nonlinear capacity). HistGBR achieved the best validation MAE and generalized well on test. The chart below summarizes validation and test MAE across models. HistGradientBoostingRegressor was selected as the primary model due to its strong performance on tabular data, built-in handling of nonlinearities, and availability within standard scikit-learn, ensuring reproducibility without external dependencies.

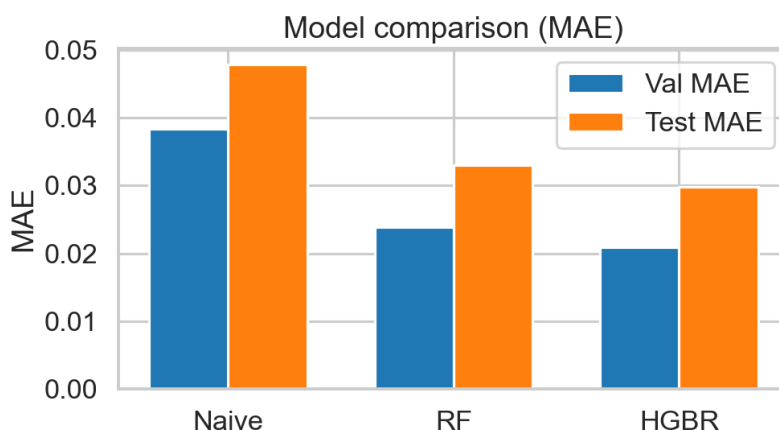


Figure 34: Model comparison (MAE) on validation vs test. HistGBR chosen as primary model.

Overall implication of EDA Overall, the EDA confirms that the dataset is internally consistent, exhibits expected diurnal and seasonal regimes, and contains well-behaved feature relationships. These findings support the validity of the modeling assumptions and justify proceeding with a unified, leakage-safe forecasting approach.

11 Model Configuration

11.1 RandomForestRegressor

- `n_estimators=200`, `max_depth=12`, `min_samples_leaf=5`
- `max_features="sqrt"`, `n_jobs=-1`, `random_state=42`

11.2 HistGradientBoostingRegressor (primary)

- `loss="squared_error"`, `learning_rate=0.05`, `max_depth=8`
- `max_iter=300`, `l2_regularization=0.0`, `random_state=42`

12 Model Testing and Deployment

12.1 Prediction Script

12.1.1 Purpose and Prerequisites

The prediction CLI produces next-hour predictions per building from a usage file (CSV/Parquet) and optional weather file. It expects hourly timestamps, building identifiers, and usage; with optional apparent temperature and precipitation for higher fidelity.

12.1.2 Workflow

1) Load the trained model and minimal preprocessing; 2) align/merge weather if provided; 3) compute leakage-safe lags per building for the last hour; 4) predict next-hour usage and write a CSV.

12.1.3 Command-Line Usage Example

```
source .venv/bin/activate && python scripts/predict.py \  
  --input data/ausgrid_with_weather_normalized.csv \  
  --region NSW \  
  --out outputs/tables/predictions_next_hour.csv
```

The pipeline is fully scripted and reproducible. Key steps:

- **Unify data:** `scripts/00_unify_datasets.py` produces a single CSV with region labels.
- **Build features:** `scripts/01_build_features.py` creates leakage-safe features and the next-hour target.
- **Train:** `scripts/02_train_models.py` performs chronological splits and trains Naive/RF/HistGBR.
- **Predict:** `scripts/predict.py` predicts next-hour usage for each building given a usage file (+ optional weather).
- **Explain:** `scripts/03_explain_segments.py` computes temperature elasticity and segments.
- **Plots:** `scripts/04_make_plots.py` + `scripts/04a_extra_plots.py` generate >30 figures.
- **Report:** `scripts/05_render_report.py` assembles this LaTeX report.

See the root `README.md` for exact commands and prediction usage examples.

12.2 Deployment Considerations

- **Internal validity:** enforce leakage controls via per-building shifts/rolling; avoid global scalers fit on full data.
- **External validity:** distribution shifts (policy changes, technology adoption) may degrade accuracy.
- **Bias:** uneven station coverage may bias elasticity; monitor subgroup metrics and confidence intervals.
- **Privacy:** anonymized identifiers; avoid re-identification; follow GDPR principles.

13 Conclusion

This project delivers an end-to-end, deployment-minded forecasting pipeline for hourly electricity usage that reflects current best practice in the prediction industry: leakage-safe data design, time-aware evaluation, and interpretable, high-performing tree ensembles. By unifying multi-region usage with weather and calendar signals, training with chronological splits, and explaining predictions via global importance and ICE/PDP, we provide both accurate next-hour forecasts and an actionable measure of temperature-driven vulnerability at the building level. The approach has limits: upstream data quality and representativeness can affect both accuracy and elasticity estimates; extreme events and sudden regime shifts remain challenging; and a single global model may underfit niche segments. Future development should explore gradient-boosting libraries with early stopping, uncertainty quantification and monitoring in production, segment-specific models where justified, and richer exogenous drivers (e.g., tariffs or mobility) to improve robustness as the system evolves. Overall, the project demonstrates the ability to design, evaluate, and interpret a realistic forecasting pipeline under practical constraints, which was the primary objective of this course assignment. Through this project, we demonstrate the ability to design a leakage-aware time-series forecasting pipeline, evaluate models under realistic deployment conditions, and interpret predictions using both global and local analysis tools. These skills align with the core learning objectives of applied data science coursework.

References

- Pedregosa et al., *Scikit-learn: Machine Learning in Python*, JMLR.
- Ke et al., *LightGBM: A Highly Efficient Gradient Boosting Decision Tree* (future work).
- Lundberg & Lee, *A Unified Approach to Interpreting Model Predictions* (future SHAP enhancements).